Ishara Connect – AI Based Sign Language Interpretation System

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Certificate

This is to certify that the students SK Nooruddin, Sumaiya Laskar, Dhiraj Kumar Mishra, MD Zaid Yasar, Sohan Mistry, and Pabakanan Bhattacharjee, students of Diploma in Computer Science and Technology department of The Calcutta Technical School, have successfully completed their Minor Project entitled ”Ishara Connect – AI Based Sign Language Interpretation System” under my supervision as part of their diploma course curriculum.

The work carried out by the students demonstrates a clear understanding of modern technologies such as Artificial Intelligence, Computer Vision, and Machine Learning. The project has been executed with sincerity, dedication, and a systematic approach. The performance of the students during the project development has been found satisfactory and meets the academic requirements prescribed by the institution.

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Acknowledgement

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We are also thankful to the Department of Computer Science and Technology of The Calcutta Technical School for providing us with the necessary infrastructure, laboratory facilities, and a conducive learning environment to carry out this project effectively.

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Table of Contents

List of Figures

List of Tables

1. Introduction

Ishara Connect is an advanced Artificial Intelligence-based system developed to interpret sign language gestures into meaningful text and speech outputs in real time. In modern society, communication plays a crucial role in personal, educational, and professional interactions. However, individuals who rely on sign language often face significant challenges when communicating with those unfamiliar with it. This project addresses this gap by introducing a technological solution that enables seamless interaction between deaf and mute individuals and the general population.

The system leverages cutting-edge computational techniques, particularly in the domains of computer vision and machine learning. By utilizing a webcam, the system captures real-time video input and processes hand movements using sophisticated algorithms. MediaPipe is employed to detect and track hand landmarks with high precision, allowing the system to extract meaningful spatial features. These features are then analyzed by a trained machine learning model capable of identifying specific gestures.

Furthermore, the system converts recognized gestures into readable text and, optionally, into audible speech output. This dual-output mechanism enhances usability in various real-world scenarios such as public communication, education, and assistive technology applications. The integration of multiple technologies ensures that the system is not only accurate but also efficient and scalable. Overall, Ishara Connect represents a significant step toward inclusive communication, demonstrating how artificial intelligence can be applied to solve real-world social challenges.

2. Motivation

Communication barriers faced by deaf and mute individuals remain a persistent issue in society. Despite advancements in technology, a large portion of the population is still unfamiliar with sign language, which limits effective interaction. This often leads to misunderstandings, reduced accessibility to services, and social isolation. The lack of a universal communication medium between sign language users and non-users highlights the need for an intelligent and accessible solution.

The motivation behind this project stems from the desire to bridge this communication gap using modern technological approaches. By developing a system capable of translating sign language gestures into text and speech, the project aims to empower individuals with hearing and speech impairments. Such a system can significantly enhance their ability to communicate in everyday situations, including educational institutions, workplaces, healthcare facilities, and public services.

Additionally, this project reflects the broader vision of leveraging Artificial Intelligence for social good. It demonstrates how machine learning and computer vision can be utilized beyond commercial applications to address real-life problems. The project also serves as a learning platform for understanding the practical implementation of AI technologies, encouraging further innovation in assistive systems. Ultimately, the motivation lies in creating a more inclusive and equitable society where communication is not a barrier but a bridge.

3. Overview of the Project

The Ishara Connect system is designed as a real-time gesture recognition and translation platform. It operates by capturing live video input through a standard webcam, making it accessible without requiring specialized hardware. The captured frames are processed using MediaPipe, a powerful framework that detects and tracks hand landmarks with remarkable accuracy. These landmarks represent key points on the hand, such as finger joints and palm positions, which are essential for gesture recognition.

Once the hand landmarks are extracted, they are passed to a machine learning model trained on a dataset of predefined gestures. The model analyzes the spatial relationships between these landmarks to classify the gesture being performed. The classification process is optimized to ensure minimal latency, enabling real-time performance.

After recognizing the gesture, the system converts it into a textual representation displayed on the screen. Additionally, a text-to-speech module can be integrated to generate audible output, further enhancing accessibility. The system combines technologies such as OpenCV for image processing, MediaPipe for feature extraction, and machine learning algorithms for classification.

The overall architecture is modular, allowing each component to function independently while contributing to the system's overall performance. This design ensures flexibility, scalability, and ease of future enhancements. The system is both efficient and user-friendly, making it suitable for practical deployment.

4. Detail Description of the Project

The Ishara Connect project is composed of several interconnected modules that work together to achieve real-time sign language recognition. The first module is the data collection module, which is responsible for gathering gesture samples. Using a webcam, multiple instances of each gesture are recorded under different conditions to ensure dataset diversity. The collected data is then stored in a structured format for further processing.

The preprocessing module plays a critical role in preparing the data for machine learning. It involves cleaning the dataset, normalizing hand landmark coordinates, and removing noise or inconsistencies. This step ensures that the model receives high-quality input, which directly impacts its performance and accuracy.

The core of the system is the machine learning module. A classification algorithm, such as Random Forest, is trained using the processed dataset. The model learns to recognize patterns in hand movements and associate them with specific gestures. During training, various parameters are optimized to achieve the best possible accuracy.

The backend of the system is developed using Flask, which provides a lightweight and efficient framework for handling real-time predictions. It manages communication between different modules and ensures smooth data flow. The frontend interface is designed to be intuitive and user-friendly, allowing users to interact with the system effortlessly. It displays the recognized text and provides visual feedback.

Overall, the integration of these modules results in a robust system capable of performing real-time gesture recognition. The architecture ensures reliability, scalability, and ease of maintenance, making it suitable for future expansion and deployment.

5. Conclusion and Scope of Further Study

The Ishara Connect project successfully demonstrates the application of Artificial Intelligence in solving a real-world problem related to communication accessibility. The system effectively recognizes sign language gestures and converts them into text and speech, thereby enabling seamless interaction between deaf and mute individuals and others. The project highlights the potential of combining computer vision and machine learning to develop assistive technologies.

The results achieved indicate that the system is both accurate and efficient, making it suitable for practical use. However, there is significant scope for further improvement and expansion. One potential enhancement is the inclusion of a larger and more diverse dataset, which would improve the model's generalization capabilities. Incorporating deep learning techniques, such as Convolutional Neural Networks (CNNs), could further enhance accuracy and performance.

Future work may also involve deploying the system on mobile devices, making it more accessible and portable. Additional features such as multilingual support, continuous sentence recognition, and integration with wearable devices can further extend its functionality. The project opens up numerous possibilities for research and development in the field of assistive technologies.

In conclusion, Ishara Connect represents a meaningful step toward inclusive communication, demonstrating how technology can be used to improve lives and create a more connected society.

6. References

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